TRIPATHI STUDY ZONE
JEE & NEET
9112409964
9075829726
CLASS 6 To 12 | CBSE | ICSE | STATE BOARDS

MOCK TEST 9

INSTRUCTIONS

1. This test will be a 3 hours Test.
2. This test consists of Physics, Chemistry and Mathematics questions with equal weightage of 100 marks.
3. Each question is of 4 marks.
4. There are three sections in the question paper consisting of Physics (Q.no.1 to 25), Chemistry (Q.no.26 to 50) and Mathematics (Q. no.51 to 75). Each section is divided into two parts, Part I consists of 20 multiple choice questions & Part II consists of 5 Numerical value type Questions.
5. There will be only one correct choice in the given four choices in Part I. For each question 4 marks will be awarded for correct choice, 1 mark will be deducted for incorrect choice for Part I Questions and zero mark will be awarded for not attempted question. For Part II Questions 4 marks will be awarded for correct answer and zero for unattempted and incorrect answer.
6. Any textual, printed or written material, mobile phones, calculator etc. is not allowed for the students appearing for the test.
7. All calculations / written work should be done in the rough sheet provided.

PHYSICS

PART-I (Multiple Choice Questions)

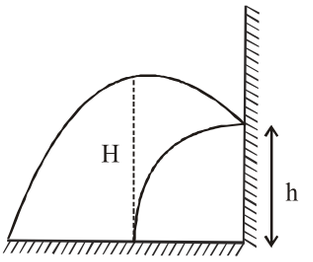
1. A river is flowing from west to east at the speed of 5 m/ minute. A man who can swim at 10 m/minute in still water is on the south bank of the river. He wants to go across the river in shortest time. How much time he will take if the width of the river is 200 m?

(a) 20 min. (b) 30 min. (c) 15 min. (d) 60 min.

2. A stuntman plans to run across a roof top and horizontally jumps on to another roof 4.9 m below the first one and at a distance of 6.2 m away. What is the minimum velocity, he must have before the jump ?

(a) 3.1 m/s (b) 4.0 m/s (c) 4.9 m/s (d) 6.2 m/s

3. A stone is projected from a horizontal plane. It attains maximum height 'H' and strikes a stationary smooth wall and falls on the ground vertically below the maximum height. Assume the collision to be elastic, the height of the point on the wall where ball will strike is –



(a) H/2 (b) H/4

(c) 3H/4 (d) None of these

4. A body falling freely from a given height H hits an inclined plane in its path at height h. As a result of this impact the direction of u becomes horizontal. For what value of h/H the body will take maxixmum time to reach the ground

(a) 1 / 2 (b) 2 / 1 (c) 1 / 1 (d) 4 / 1

5. A satellite is revolving round the earth with velocity v. The minimum percentage increase in its velocity necessary for the escape of satellite will be

(a) 100% (b) 50% (c) 82.3% (d) 41.4%

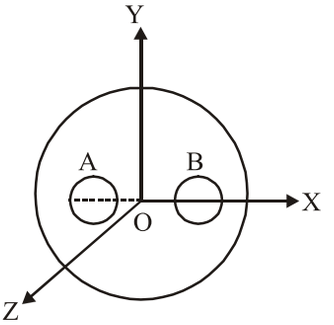
Space for Rough Work

6. From a vertical water jet placed on the ground, water drops come out with same velocity at different angles. On landing if they jump with $1/4^{th}$ of the original velocity and stop, the maximum distance upto which water can reach is

(a) v^2/g (b) v^4/g^2

(c) $(17/16)\,v^2/g$ (d) None of these

7. A solid sphere of uniform density and radius 4 units is located with its centre at the origin O of coordinates.



Two spheres of equal radii 1 unit, with their centres at A (–2, 0, 0) and B (2, 0, 0) respectively, are taken out of the solid, leaving behind spherical cavities as shown in figure, then

(a) the gravitational force due to this object at the origin is zero

(b) the gravitational force at point B(2, 0, 0) is zero

(c) the gravitational potential is same at all points of the circle $y^2 + z^2 = 36$

(d) both (a) and (c)

8. The rectangular surface of area 8 cm × 4 cm of a black body at temperature 127°C emits energy E per second. If the length and breadth are reduced to half of the initial value and the temperature is raised to 327°C, the rate of emission of energy becomes

(a) $\frac{3}{8}E$ (b) $\frac{81}{16}E$ (c) $\frac{9}{16}E$ (d) $\frac{81}{64}E$

9. Inside a cylinder closed at both ends is a movable piston. On one side of the piston is a mass m of a gas, and on the other side a mass 2m of the same gas. What fraction of volume of the cylinder will be occupied by the larger mass of the gas when the piston is in equilibrium? The temperature is the same throughout.

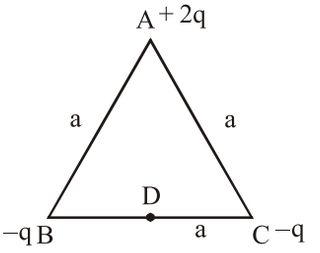
(a) 1/4 (b) 1/2 (c) 2/3 (d) 1/3

10. Two identical containers A & B with frictionless pistons contain the same ideal gas at the same temperature & the same volume V. The mass of the gas in A is m_A & that in B is m_B. The gas in each cylinder is now allowed to expand isothermally to the same final volume 2V. The changes in the pressure in A & B are found to be ΔP & 1.5 ΔP respectively. Then

(a) $4\,m_A = 9\,m_B$ (b) $2m_A = 3\,m_B$

(c) $3\,m_A = 2\,m_B$ (d) $9m_A = 4m_B$

11. Three charges of (+ 2q), (– q) and (– q) are placed at the corners A, B and C of the equilateral triangle of side a as shown in the given figure. Then the dipole moment of this combination is



(a) qa (b) zero

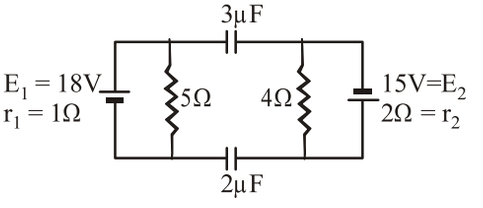
(c) qa $\sqrt{3}$ (d) $\frac{2}{\sqrt{3}}$ qa

12. We have a parallel plate capacitor of capacitance C, potential difference V and plate separation d. Someone pulls the plate apart such that the plate separation becomes 2d. The amount of work done is (given m = mass of either plate)

(a) mgd (b) $\frac{1}{4}CV^2$

(c) $\left(\frac{3}{4}\right)CV^2$ (d) $(1/9)CV^2$

13. Two batteries, two resistors and two condensers are connected as shown. The charge on 2μF capacitor is



(a) 30 μC (b) 20 μC (c) 25 μC (d) 48 μC

14. Torques τ_1 and τ_2 are required for a magnetic needle to remain perpendicular to the magnetic fields at two different places. The magnetic fields at the those places are B_1 and B_2 respectively; then ratio $\frac{B_1}{B_2}$ is

(a) $\frac{\tau_2}{\tau_1}$ (b) $\frac{\tau_1}{\tau_2}$ (c) $\frac{\tau_1+\tau_2}{\tau_1-\tau_2}$ (d) $\frac{\tau_1-\tau_2}{\tau_1+\tau_2}$

15. In an A.C. circuit with voltage V and current I the power dissipated

(a) depends on the phase between V and I

(b) $\frac{1}{\sqrt{2}}$VI

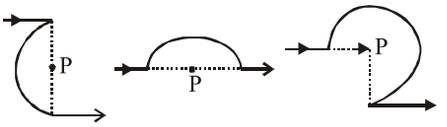
(c) $\frac{1}{2}$VI

(d) VI

16. A direct current of 2A and an alternating current having a maximum value of 2A flow through two identical resistances. The ratio of heat produced in the two resistances will be

(a) 1 : 1 (b) 1 : 2 (c) 2 : 1 (d) 4 : 1

17. The magnetic field at the centre of a circular coil of radius r is p times that due to a long straight wire at a distance r from it, for equal currents. The following diagram shows three cases : in all cases the circular part has radius r and straight ones are infinitely long. For the same current, the magnetic field at centres P in case 1, 2, 3 have the ratio



(a) $\left(\frac{-\pi}{2}\right):\left(\frac{\pi}{2}\right):\left(\frac{3\pi}{4}-\frac{1}{2}\right)$

(b) $\left(\frac{-\pi}{2}+1\right):\left(\frac{\pi}{2}+1\right):\left(\frac{3\pi}{4}+\frac{1}{2}\right)$

(c) $\frac{-\pi}{2}:\frac{\pi}{2}:\frac{3\pi}{4}$

(d) $\left(\frac{-\pi}{2}-1\right):\left(\frac{\pi}{2}-\frac{1}{4}\right):\left(\frac{3\pi}{4}+\frac{1}{2}\right)$

18. A particle of mass m is under a potential $v(x) = (1/2)k_1x^2$ for $x > 0$ and $v(x) = k_1x^2$ for $x < 0$. When disturbed a little from the position $x = 0$, it will

(a) not execute SHM

(b) execute SHM with $T = 2\pi\sqrt{\frac{m}{3k_1}}$

(c) execute SHM with $T^2 = 2\pi^2 m/(k_1^2)$

(d) execute SHM with $T = \frac{\pi\sqrt{m}}{\sqrt{k_1\left(1+\frac{1}{\sqrt{2}}\right)}}$

19. A far sighted person can see clearly object beyond 100 cm distance. If he wants to see clearly an object at 40 cm distance, then the power of the lens he shall require is

(a) +1.5 D (b) –1.5D (c) +3.0 D (d) –3.0 D

20. A hydrogen atom emits green light when it changes from n = 4 energy level to the n = 2 level. Which colour of light would the atom emit when it changes from n = 5 level to the n=2 level?

(a) red (b) yellow (c) green (d) violet

PART-II (Numerical Answer Questions)

21. A 20 cm long capillary tube is dipped in water. The water rises up to 8 cm. If the entire arrangement is put in a freely falling elevator the length (in cm) of water column in the capillary tube will be

22. A particle of charge 16×10^{-16} C moving with velocity 10 ms^{-1} along *x*-axis enters a region where magnetic field of induction $\vec{B}$ is along the *y*-axis and an electric field of magnitude 10^4 Vm^{-1} is along the negative *z*-axis. If the charged particle continues moving along *x*-axis, the magnitude of $\vec{B}$ (in Wb m^{-2}) is :

23. How many degree of freedom the gas molecules have if under STP the gas density ρ=1.3 kg/m^3 and the velocity of sound propagation in it is 330 m/sec ?

24. A thin sheet of glass (μ = 1.5) of thickness 6 microns introduced in the path of one of interfering beams in a double slit experiment shifts the central fringe to a position previously occupied by fifth bright fringe. Then the wave length (in Å) of the light used is

25. The work function of a metallic surface is 5.01 eV, photoelectrons are emitted when light of wavelength 2000Å falls on it. The minimum potential difference (in volts) required to stop the fastest photoelectrons ($h = 4.14 \times 10^{-15}$ eV–s)

CHEMISTRY

PART-I (Multiple Choice Questions)

26. The tendency of a reaction to take place spontaneously is greatest when the reaction is

(a) Exothermic and randomness increases

(b) Endothermic & randomness increases

(c) Exothermic and randomness decreases

(d) Endothermic & randomness decreases

27. Which of the following orders regarding the bond order is correct ?

(a) $O_2^- > O_2 > O_2^+$ (b) $O_2^- < O_2 < O_2^+$

(c) $O_2^- > O_2 < O_2^+$ (d) $O_2^- < O_2 > O_2^+$

28. In order to convert R – X to R – R' which reaction is most suitable

(a) Corey House reaction (b) Kolbe's reaction

(c) Williamson's synthesis (d) Wurtz reaction

29. Which of the following order of root mean square speed of different gases at same temperature is true?

(a) $(u_{rms})_{H_2} > (u_{rms})_{CH_4} > (u_{rms})_{NH_3} > (u_{rms})_{CO_2}$

(b) $(u_{rms})_{H_2} < (u_{rms})_{CH_4} < (u_{rms})_{NH_3} < (u_{rms})_{CO_2}$

(c) $(u_{rms})_{H_2} < (u_{rms})_{CH_4} > (u_{rms})_{NH_3} > (u_{rms})_{CO_2}$

(d) $(u_{rms})_{H_2} > (u_{rms})_{CH_4} < (u_{rms})_{NH_3} < (u_{rms})_{CO_2}$

30. Positive Beilstein test for halogens shows that

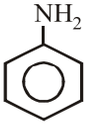
(a) a halogen is definitely present

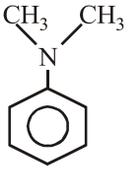
(b) a halogen may be present

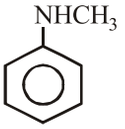
(c) a halogen is absent

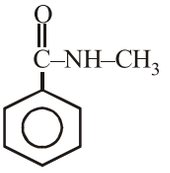
(d) none of these.

31. Which of the following gives n-nitrosoamine on reaction with nitrous acid?

(a) 

(b) 

(c) 

(d) 

32. Consider the reaction

$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$ in a closed container at equilibrium. What would be the effect of addition of $CaCO_3$ on the equilibrium concentration of CO_2?

(a) Increases (b) Decreases

(c) Data insufficient (d) Remains unaffected

33. To prepare 3-ethylpentane-3-ol the reagents needed are

(a) $CH_3CH_2MgBr + CH_3COCH_2CH_3$

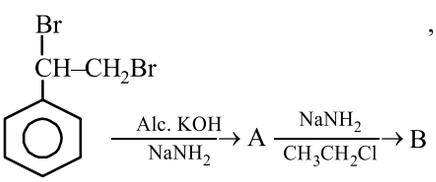
(b) $CH_3MgBr + CH_3CH_2CH_2COCH_2CH_3$

(c) $CH_3CH_2MgBr + CH_3CH_2COCH_2CH_3$

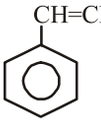
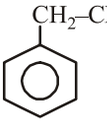
(d) $CH_3CH_2CH_2MgBr + CH_3COCH_2CH_3$

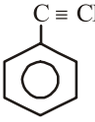
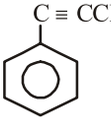
34. How many P–H and O–H bonds respectively, are present in $H_4P_2O_7$ molecule?

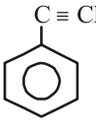
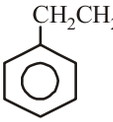
(a) 1 and 3 (b) 0 and 4 (c) 4 and 0 (d) 2 and 3

35. 

A and B are

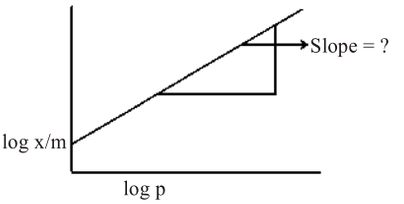
(a) A =  ; B = 

(b) A =  ; B = 

(c) A =  ; B = 

(d) None of these

36. In the Freundlich adsorption isotherm equation graph. The slope is equal to



(a) n (b) 1/n

(c) log k (d) log k – 1/n

37. Which one of the following on treatment with 50% aqueous sodium hydroxide yields the corresponding alcohol and acid?

(a) C_6H_5CHO (b) $CH_3CH_2CH_2CHO$

(c) $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ (d) $C_6H_5CH_2CHO$

38. The complexes of Nickel (II) can be

(a) Square planar, tetrahedral and octahedral

(b) Square planar and octahedral

(c) Tetrahedral and octahedral

(d) Square planar only

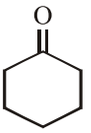
39. Which of the following represents the correct order of stability?

(a) $H-C\equiv C:^- > H_2C=CH:^- > H_3C-CH_2:^-$

(b) $H_3C-CH_2:^- > H_2C=CH:^-\ H-C\equiv C:^-$

(c) $H_3C-CH_2:^- = H_2C=CH:^- = H-C\equiv C:^-$

(d) None of these

40.  $\xrightarrow{NH_2OH}$ A $\xrightarrow{H_2SO_4}$ B. The product B is

(a) Caprolactam (b) Lactone
(c) Perlone (d) Nylon-6

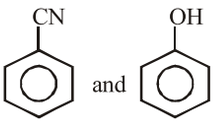
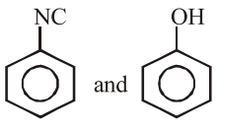
41. During steam distillation of a mixture of *o*-nitrophenol and *p*-nitrophenol
(a) vapours of *o*-nitrophenol are carried by steam because of its lower boiling point due to chelation.
(b) vapours of *o*-nitrophenols are carried by steam because of its lower boiling point and solubility in steam.
(c) vapours of *o*-nitrophenol are carried by steam because its boiling point is reduced by steam.
(d) vapours of *p*-nitrophenol are carried by steam because of its lower boiling point.

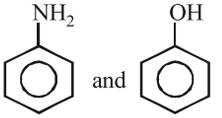
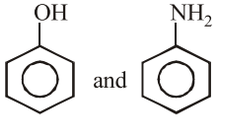
42. The hybridization states of the central atoms in the complexes $[Fe(CN)_6]^{3-}$, $[Fe(CN)_6]^{4-}$ and $[Co(NO_2)_6]^{3-}$ are
(a) d^2sp^3, sp^3 and d^4s^2 respectively
(b) d^2sp^3, sp^3d and sp^3d^2 respectively
(c) d^2sp^3, sp^3d^2 and dsp^2 respectively
(d) all d^2sp^3

43. Among the acids given below
CH_3CH_2COOH (X); $CH_2 = CHCOOH$ (Y) and $CH \equiv C\text{-}COOH$ (Z)
The correct order of increasing acid strength is
(a) X < Y < Z (b) X < Z < Y (c) Y < X < Z (d) Z < Y < X

44. Which of the following is most basic in nature
(a) $Lu(OH)_3$ (b) $Nd(OH)_3$
(c) $Ce(OH)_3$ (d) all are acidic in nature

45. A mixture of two aromatic compounds A and B is separated by dissolving in chloroform followed by extraction with aq. KOH solution. The alkaline aqueous layer gives a mixture of two isomeric compounds on treatment with carbon tetrachloride. The organic layer containing compound A gives an unpleasant odour on treatment with alcoholic solution of KOH. Compounds A and B respectively are

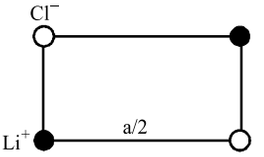
(a)  (b) 

(c)  (d) 

PART-II (Numerical Answer Questions)

46. The threshold frequency of a metal is $1 \times 10^{15}\ s^{-1}$. The ratio of maximum kinetic energies of the photoelectrons when the metal is irradiated with radiations of frequencies $1.5 \times 10^{15}\ s^{-1}$ and $2 \times 10^{15}\ s^{-1}$ respectively would be

47. Lithium chloride has a cubic structure as shown below. If the edge length is 400 pm., then the radii of Cl^- ions is



48. Normality of a mixed solution of sulphuric acid and hydrochloric acid is 0.6 N. 20 ml of this solution gives 0.4305 g of AgCl on reacting with $AgNO_3$ solution. The strength of H_2SO_4 in g/*l* in the mixed solution is

49. The halflife of a first order reaction is 10 min. If initial amount is 0.08 mol L^{-1} and concentration at some instant is 0.01 mol L^{-1}. What is the time elapsed ?

50. The molar specific heat of air at room temperature and 1atm pressure is 25 $J\ K^{-1}mol^{-1}$. How much heat is required (in kJ) to heat the room through 10 Kelvin at room temperature if 144g of air is present in room. Vapour density of air = 14.4.

MATHEMATICS

PART-I (Multiple Choice Questions)

51. If $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = -3\hat{i} + \hat{j} - \hat{k}$ and
$\vec{r} \times \vec{a} = \vec{b} \times \vec{a}, \vec{r} \times \vec{b} = \vec{a} \times \vec{b}$
then a unit vector in the direction of $\vec{r}$ is
(a) $\frac{1}{3}(-2\hat{i} + \hat{j} - \hat{k})$ (b) $\frac{1}{3}(-2\hat{i} - \hat{j} + 2\hat{k})$
(c) $\frac{1}{3}(-2\hat{i} - \hat{j} - 2\hat{k})$ (d) None of these

52. If $P = \{ x \in \mathbf{R} : f(x) = 0 \}$ and
$Q = \{ x \in \mathbf{R} : g(x) = 0 \}$ then $P \cup Q$ is
(a) $\{ x \in \mathbf{R} : f(x) + g(x) = 0 \}$
(b) $\{ x \in \mathbf{R} : f(x)\, g(x) = 0 \}$
(c) $\{ x \in \mathbf{R} : (f(x))^2 + (g(x))^2 = 0 \}$
(d) none of these

53. If $(\vec{a}\times\vec{b})\times(\vec{c}\times\vec{d})+(\vec{a}\times\vec{c})\times(\vec{d}\times\vec{b})+(\vec{a}\times\vec{d})\times(\vec{b}\times\vec{c})=\lambda a$, where λ is a scalar then the value of λ is

(a) zero (b) $-2[\vec{b}\,\vec{c}\,\vec{d}]$

(c) $2[\vec{a}\,\vec{b}\,\vec{c}]$ (d) $[\vec{b}\,\vec{c}\,\vec{d}]$

54. $\int e^{(\log x+ax^2)}\cos(bx^2+c)\,dx$ is equal to

(a) $\frac{1}{\sqrt{a^2+b^2}}e^{ax^2}\cos(bx^2+c+\tan^{-1}\frac{b}{a})+A$

(b) $\frac{1}{2\sqrt{a^2+b^2}}e^{ax^2}\cos(bx\ -c-\tan^{-1}\frac{b}{a})+A$

(c) $\frac{1}{\sqrt{a^2+b^2}}e^{ax^2}\cos(bx\ +c-\tan^{-1}\frac{b}{a})+A$

(d) $\frac{1}{2\sqrt{a^2+b^2}}e^{ax^2}\cos(bx\ +c-\tan^{-1}\frac{b}{a})+A$

55. If $y^2=P(x)$ is a polynomial of degree 3, then

$2\frac{d}{dx}\left[y^3\frac{d^2y}{dx^2}\right]$ is equal to

(a) $P(x)+P'(x)$ (b) $P(x)+P'''(x)$

(c) $P(x)\,P'''(x)$ (d) None of these

56. Which of the following is an even function?

(a) $f(x)=\log\left(\frac{1-x}{1+x}\right)$ (b) $f(x)=\log(x+\sqrt{1+x^2})$

(c) $f(x)=\frac{x}{e^x-1}+\frac{x}{2}+1$ (d) $f(x)=e^{2x}+\sin x$

57. A plane meets the coordinate axes in points A, B, C and the centroid of the triangle ABC is (α, β, γ). The equation of the plane is

(a) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=3$ (b) $\alpha x+\beta y+\gamma z=3\alpha\beta\gamma$

(c) $\frac{x}{\alpha}+\frac{y}{\beta}+\frac{z}{\gamma}=\frac{1}{2}$ (d) None of these

58. The centre of the ellipse $\frac{(x+y-2)^2}{16}+\frac{(x-y)^2}{9}=1$ is :

(a) $(0,0)$ (b) $(1,1)$

(c) $(1,0)$ (d) $(0,1)$

59. The point (2a, a) lies inside the region bounded by the parabola $x^2=4y$ and its latus rectum. Then,

(a) $0\le a\le 1$ (b) $0<a<1$

(c) $a>1$ (d) $a<0$

60. If ΔABC is right angled at A then the inradius of the triangle is

(a) $2(a+b-2)$ (b) $2(b+c-a)$

(c) $\frac{b+c-a}{2}$ (d) $\frac{b+c-a}{4}$

61. If $\frac{1}{\sqrt{b}+\sqrt{c}},\frac{1}{\sqrt{c}+\sqrt{a}},\frac{1}{\sqrt{a}+\sqrt{b}}$ are in A.P. then $9^{ax+1},9^{bx+1},9^{cx+1}$, $x\neq 0$ are in :

(a) G.P. (b) G.P. only if $x<0$

(c) G.P. only if $x>0$ (d) none of these

62. If $a_k=\int_0^{\pi}\frac{\sin(2k-1)x}{\sin x}dx$, then –

(a) $a_1, a_2, \ldots$ are in A.P.

(b) $a_1, a_2, \ldots$ are in G.P.

(c) $a_1, a_2, \ldots$ are in H.P.

(d) $a_1, a_2, \ldots$ form a constant sequence

63. If the roots of $a(b-c)x^2+b(c-a)x+c(a-b)=0$ are equal, then a, b, c are in :

(a) A.P. (b) G.P.

(c) H.P. (d) none of these

64. If z_1, z_2, z_3, z_4 be the vertices of a quadrilateral taken in order such that $z_1+z_3=z_2+z_4$ and

$|z_1-z_3|=|z_2-z_4|$ then $\arg\left(\frac{z_1-z_2}{z_3-z_2}\right)=$

(a) $\frac{\pi}{2}$ (b) $\pm\frac{\pi}{2}$ (c) $\frac{\pi}{3}$ (d) $\frac{\pi}{6}$

65. Let $f(x)=e^x$, $g(x)=\sin^{-1}x$ and $h(x)=f(g(x))$, then $h'(x)/h(x)=$

(a) $e^{\sin^{-1}x}$ (b) $1/\sqrt{1-x^2}$

(c) $\sin^{-1}x$ (d) $1/(1-x^2)$

Space for Rough Work

66. Which of the following functions is differentiable at $x = 0$?

(a) $\cos(|x|) + |x|$ (b) $\cos(|x|) - |x|$

(c) $\sin(|x|) + |x|$ (d) $\sin(|x|) - |x|$

67. Let $f : R \to R$ be function defined by $f(x) = \sin(2x - 3)$, then f is

(a) injective (b) surjective

(c) bijective (d) None of these

68. Let E = (1, 2, 3, 4) and F = (1, 2). Then the number of onto functions from E to F is

(a) 14 (b) 16 (c) 12 (d) 8

69. Which of the following functions is NOT one-one ?

(a) $f : R \to R$ defined by $f(x) = 6x - 1$.

(b) $f : R \to R$ defined by $f(x) = x^2 + 7$.

(c) $f : R \to R$ defined by $f(x) = x^3$.

(d) $f : R - \{7\} \to R$ *defined by* $f(x) = \dfrac{2x+1}{x-7}$.

70. $\lim\limits_{n\to\infty}\left[\dfrac{1}{n^2}\sec^2\dfrac{1}{n^2} + \dfrac{2}{n^2}\sec^2\dfrac{4}{n^2} \ldots\ldots\ldots + \dfrac{1}{n}\sec^2 1\right]$ equals

(a) $\dfrac{1}{2}\sec 1$ (b) $\dfrac{1}{2}\operatorname{cosec} 1$

(c) $\tan 1$ (d) $\dfrac{1}{2}\tan 1$

PART-II (Numerical Answer Questions)

71. In a two player game that always has a winner, A beats B with probability 2/3; B beats C with probability 2/3 & C beats A with the same probability. If B plays with C and then the winner plays with A, the chance that A will be the final winner is

72. If $\vec{a} = \vec{i} + \vec{j} + \vec{k}$, $\vec{b} = 4\vec{i} + 3\vec{j} + 4\vec{k}$ and $\vec{c} = \vec{i} + \alpha\vec{j} + \beta\vec{k}$ are linearly independent vectors and $|\vec{c}| = \sqrt{3}$, then $\alpha^2 + \beta =$

73. If the area lying between the curves $y = \tan x$, $y = \cot x$ and x-axis, $x \in \left[0, \dfrac{\pi}{2}\right]$ is a log b, then $\dfrac{a}{b} =$

74. If the eccentricity of the hyperbola $x^2 - 3y^2 = 2x + 8$ is $\dfrac{a}{\sqrt{b}}$, then $\dfrac{b}{a} =$

75. The function defined by $f(x) = \begin{cases} \left(x^2 + e^{\frac{1}{2-x}}\right)^{-1}, & x \neq 2, \text{ is} \\ k & x = 2 \end{cases}$

continuous from right at the point x =2, then k is equal to

RESPONSE SHEET

PHYSICS

1.	ⓐⓑⓒⓓ	2.	ⓐⓑⓒⓓ	3.	ⓐⓑⓒⓓ	4.	ⓐⓑⓒⓓ	5.	ⓐⓑⓒⓓ
6.	ⓐⓑⓒⓓ	7.	ⓐⓑⓒⓓ	8.	ⓐⓑⓒⓓ	9.	ⓐⓑⓒⓓ	10.	ⓐⓑⓒⓓ
11.	ⓐⓑⓒⓓ	12.	ⓐⓑⓒⓓ	13.	ⓐⓑⓒⓓ	14.	ⓐⓑⓒⓓ	15.	ⓐⓑⓒⓓ
16.	ⓐⓑⓒⓓ	17.	ⓐⓑⓒⓓ	18.	ⓐⓑⓒⓓ	19.	ⓐⓑⓒⓓ	20.	ⓐⓑⓒⓓ
21.		22.		23.		24.		25.	

CHEMISTRY

26.	ⓐⓑⓒⓓ	27.	ⓐⓑⓒⓓ	28.	ⓐⓑⓒⓓ	29.	ⓐⓑⓒⓓ	30.	ⓐⓑⓒⓓ
31.	ⓐⓑⓒⓓ	32.	ⓐⓑⓒⓓ	33.	ⓐⓑⓒⓓ	34.	ⓐⓑⓒⓓ	35.	ⓐⓑⓒⓓ
36.	ⓐⓑⓒⓓ	37.	ⓐⓑⓒⓓ	38.	ⓐⓑⓒⓓ	39.	ⓐⓑⓒⓓ	40.	ⓐⓑⓒⓓ
41.	ⓐⓑⓒⓓ	42.	ⓐⓑⓒⓓ	43.	ⓐⓑⓒⓓ	44.	ⓐⓑⓒⓓ	45.	ⓐⓑⓒⓓ
46.		47.		48.		49.		50.	

MATHEMATICS

51.	ⓐⓑⓒⓓ	52.	ⓐⓑⓒⓓ	53.	ⓐⓑⓒⓓ	54.	ⓐⓑⓒⓓ	55.	ⓐⓑⓒⓓ
56.	ⓐⓑⓒⓓ	57.	ⓐⓑⓒⓓ	58.	ⓐⓑⓒⓓ	59.	ⓐⓑⓒⓓ	60.	ⓐⓑⓒⓓ
61.	ⓐⓑⓒⓓ	62.	ⓐⓑⓒⓓ	63.	ⓐⓑⓒⓓ	64.	ⓐⓑⓒⓓ	65.	ⓐⓑⓒⓓ
66.	ⓐⓑⓒⓓ	67.	ⓐⓑⓒⓓ	68.	ⓐⓑⓒⓓ	69.	ⓐⓑⓒⓓ	70.	ⓐⓑⓒⓓ
71.		72.		73.		74.		75.	